



University of Sri Jayewardenepura
Faculty of Applied Science
Department of Mathematics
BSc Degree First Year First Semester Course Unit Examination
2026 July/August

PMT 1013 Foundations of Mathematics

Answer **all (05)** Questions

Time: 03 hours

$\mathbb{Z}$ denotes the set of all integers.

01. (a) Prove the following statements:

- (i) For all integers $a, b, c \in \mathbb{Z}$, if $a|b$ and $a|c$, then $a|(b + c)$.
- (ii) For all integers $a, b, c \in \mathbb{Z}$, if $a|b$ and $a \nmid c$, then $a \nmid (b + c)$.
- (iii) For all integers $a, b \in \mathbb{Z}$, if $a \nmid -b$, then $a \nmid b$.

(b) Express the following argument in symbolic form and, **without** using a truth table, determine whether it is valid or invalid.

If integer is divisible by 4, then it is even.

If integer is even and perfect square, then it is divisible by 4.

16 is even.

Therefore, 16 is a perfect square.

[20 Marks]

02. (a)(i) Let $x_1 = 1$ and $x_{n+1} = \sqrt{1 + 2x_n}$ for all $n \in \mathbb{N}$. Using Mathematical Induction prove that for all $n \in \mathbb{N}$, $x_n < 4$.

(ii) Let $x_1 = 1, x_2 = 1$ and $x_n = x_{n-1} + x_{n-2}$ for all $n \geq 3$. Use Strong Mathematical Induction to prove that for all $n \in \mathbb{N}$, $x_n = \frac{1}{\sqrt{5}} \left[\left(\frac{1+\sqrt{5}}{2} \right)^n - \left(\frac{1-\sqrt{5}}{2} \right)^n \right]$.

(b) Let A, B and C be three non-empty subsets of the universal set E .

- (i) Use definitions to prove that $(A \setminus B) \setminus C = A \setminus (B \cup C)$ and $(A \setminus B) \cap (B \setminus C) = \emptyset$.
- (ii) Prove or disprove $A \setminus (B \cup C) = (A \setminus B) \cup (A \setminus C)$.

[20 Marks]

03. (a) Let A, B and C be three non-empty subsets of the universal set E . Prove that:

- (i) $A \times (B \setminus C) = (A \times B) \setminus (A \times C)$.
- (ii) $B \subseteq C \Rightarrow A \times B \subseteq A \times C$.

(b) For each $n \in \mathbb{Z}$, define $A_n = \{k | k \in \mathbb{N} \text{ and } k > n\}$. Determine whether the following statements are true or false. Justify your answer.

- (i) $A_1 = \mathbb{N}$.
- (ii) For all $i, j \in \mathbb{N}$, if $i < j$, then $A_i \subseteq A_j$.
- (iii) $\bigcap_{i \in \mathbb{N}} A_i = \emptyset$.

(c) Let Λ be a non-empty indexing set, and $\mathcal{A} = \{A_\alpha | \alpha \in \Lambda\}$ be an indexing family of sets, where A_α is a subset of the universal set E for each $\alpha \in \Lambda$ and B be a any subset of E .

(i) Prove that

$$B \cup \bigcap_{\alpha \in \Lambda} A_\alpha = \bigcap_{\alpha \in \Lambda} (B \cup A_\alpha).$$

(ii) Show that

$$B \cap \bigcup_{\alpha \in \Lambda} A_\alpha = \bigcup_{\alpha \in \Lambda} (B \cap A_\alpha).$$

[20 Marks]

04. (a) Let $\mathcal{R}$ and $\mathcal{S}$ be relations on a set A . Prove that:

- (i) $(\mathcal{R}^{-1})^{-1} = \mathcal{R}$. (ii) $\mathcal{R} \subseteq \mathcal{S} \Rightarrow \mathcal{R}^{-1} \subseteq \mathcal{S}^{-1}$. (iii) $(\mathcal{R} \cup \mathcal{S})^{-1} = \mathcal{R}^{-1} \cup \mathcal{S}^{-1}$.

(b) Let $\mathcal{R}$ be a relation on a set A . Prove that:

- (i) $\mathcal{R}$ is transitive if and only if $\mathcal{R} \circ \mathcal{R} \subseteq \mathcal{R}$.
- (ii) $\mathcal{R}$ is antisymmetric if and only if $\mathcal{R} \cap \mathcal{R}^{-1} \subseteq I_A$.
- (iii) If $A = \{1, 2, 3, 4\}$ and $\mathcal{R} = \{(1, 3), (1, 4), (3, 2), (3, 3), (3, 4)\}$, then determine whether $\mathcal{R}$ is transitive? $\mathcal{R}$ is antisymmetric?, and partial order? If the given relation is not a partial order, add the minimum number of ordered pairs necessary so that it becomes a partial order.

(c) Let A be a non-empty set and “ $\sim$ ” an equivalence relation on A . Prove that:

- (i) $x \sim y \Leftrightarrow [x] = [y]$.
- (ii) $A = \bigcup_{x \in A} [x]$.

(d) Relation ρ on define on $\mathbb{Z}$ by $x \rho y$ if and only if $3 | x - y$.

- (i) Show that ρ is an equivalence relation on $\mathbb{Z}$.
- (ii) Find the equivalence classes $[0], [1], [2]$.
- (iii) Show that $\{[0], [1], [2]\}$ is a partition of $\mathbb{Z}$.

[20 Marks]

05. (a) Let $f: A \rightarrow B$ be a function from non-empty sets A into B . Let U be a subset of A and V is a subset of B . Define:

- (i) $f(U)$ the image of U under f .
- (ii) $f^{-1}(V)$ the inverse image of V under f .

(b) Let $f: \mathbb{R} \setminus \{0\} \rightarrow \mathbb{R}$ defined by $f(x) = x + \frac{1}{x}$ for all $x \in \mathbb{R} \setminus \{0\}$.

- (i) Find $f(U)$, where $U = \{x | x \in \mathbb{R} \text{ and } 1 < x \leq 3\}$.
- (ii) Find $f^{-1}(V)$, where $V = \{x | x \in \mathbb{R} \text{ and } -2 < x < 0\}$.
- (iii) Show that f is injective. Is f surjective? Justify your answer.

(c) Let $f: A \rightarrow B$ be a function from non-empty sets A into B . Let E and F be subsets of A and G and H be subsets of B . Prove that:

- (i) $f(E \cap F) \subseteq f(E) \cap f(F)$.
- (ii) $f^{-1}(G \cap H) \subseteq f^{-1}(G) \cap f^{-1}(H)$.

(d) Suppose $f: A \rightarrow B$ and $g: B \rightarrow C$ are functions. Prove that

- (i) If $g \circ f$ is onto, then, g is onto.
- (ii) If $g \circ f$ is one-to-one, then f is one-to-one.

[20 Marks]
